Investigation: PETRÓLEO BRASILEIRO S.A. - PETROBRAS (CVM code 9512), fiscal year 2010, as it was on 2011-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Petróleo e Gás (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2010 DFP, version 3, filed on 2011-03-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2011-02-25); 2 (2011-03-04); 3 (2011-03-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=5169

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.944 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 79 (n = 317)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.057 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 67 (n = 406)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.120 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 84 (n = 414)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.795 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value
- Percentile in group and year: 95 (n = 416)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.166 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: fewer than 10 companies with a computable value

Percentile in group and year: 56 (n = 414)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 1.197 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 79 (n = 327)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.967 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 50 (n = 411)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.034 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 48 (n = 432)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.755 (index)
M-score component; it has no rule of its own
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 8 (n = 424)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −11.6 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 16.9 days)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 20 (n = 391)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −6.79 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 6 (n = 383)
The company's own history: fewer than 5 previous fiscal years with a value

Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.040 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: fewer than 10 companies with a computable value
Percentile in group and year: 45 (n = 428)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

Financial risk (context only)

Altman Z'' (financial risk, context only)
Value: 3.406 (index)
Components: X1 = 0.096; X2 = 0.195; X3 = 0.088; X4 = 1.479 (indices)
Percentile in sector and year: 70 (n = 533); percentile in the non-financial universe: the sector has fewer than 10 companies with a value
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z''
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (5 computable)
Signals:
- Return on assets (ROA): 1
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): —
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): —
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): —
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- FCA 2010, v1: 2010-05-31
- FCA 2010, v2: 2010-08-02
- DFP 2010, v1: 2011-02-25
- DFP 2010, v2: 2011-03-04
- DFP 2010, v3: 2011-03-04

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentamadequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da PetróleoBrasileiro S.A. - Petrobras em 31 de dezembro de 2010, o desempenho de suas operações e osseus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeisadotadas no Brasil.Opinião sobre as demonstrações contábeis consolidadas7."
Auditor in the 2010 FCA: KPMG Auditores Independentes.
Auditor changes: none in the FCAs filed by the date.
Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-10T19:30:29+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.